Experiment 2a. The blind spot (1)

(a) Close your left eye.

(b) Hold this page about 60 cm away from your face.

(c) Concentrate on the cross with the right eye and slowly bring the page closer to the face, still concentrating on the cross.

(d) Repeat the experiment with the page held sideways so that the dot is above the cross; and also with the page upside down with the dot on the left of the cross.

(e) Describe what you see in each case and explain it as far as possible.



Experiment 2b. The blind spot (2)

(a) Close the left eye and hold the page at arm's length.

(b) With the right eye, concentrate on the cross and bring the page slowly nearer your face. At first, the short line on the right will disappear because its image falls on the blind spot.

(c) When the short line reappears, the image of the gap will be on the blind spot. Describe what you see at this point.



SOLUTIONS

Experiment 2a. The blind spot (1) - *discussion - answers*

The act of concentrating attention on the cross throws its image on to the fovea and the image of the dot on to the periphery of the retina to the left of the fovea. As the book approaches the face, the distance between the images of the cross and dot increases until that of the dot falls on the blind spot, whereupon the dot disappears (Figs. 1a and b).
These results can be used to establish the relative positions of the blind spot and fovea since the dot cannot be made to disappear when the experiment is repeated with the dot and cross in other positions.
One is not usually aware of the blind spot because (a) both eyes are used, (b) the eyes make constant scanning movements (c) the gap in the visual field never coincides with the image of the object on which we are concentrating and (d) the brain 'fills in' the gap (see Experiment 2b).

Experiment 2b. The blind spot (2) - *discussion - answers*

When the image of the gap falls on the blind spot, the line appears to be continuous. It is as if the brain does not normally recognize the absence of sensory information from the blind spot.

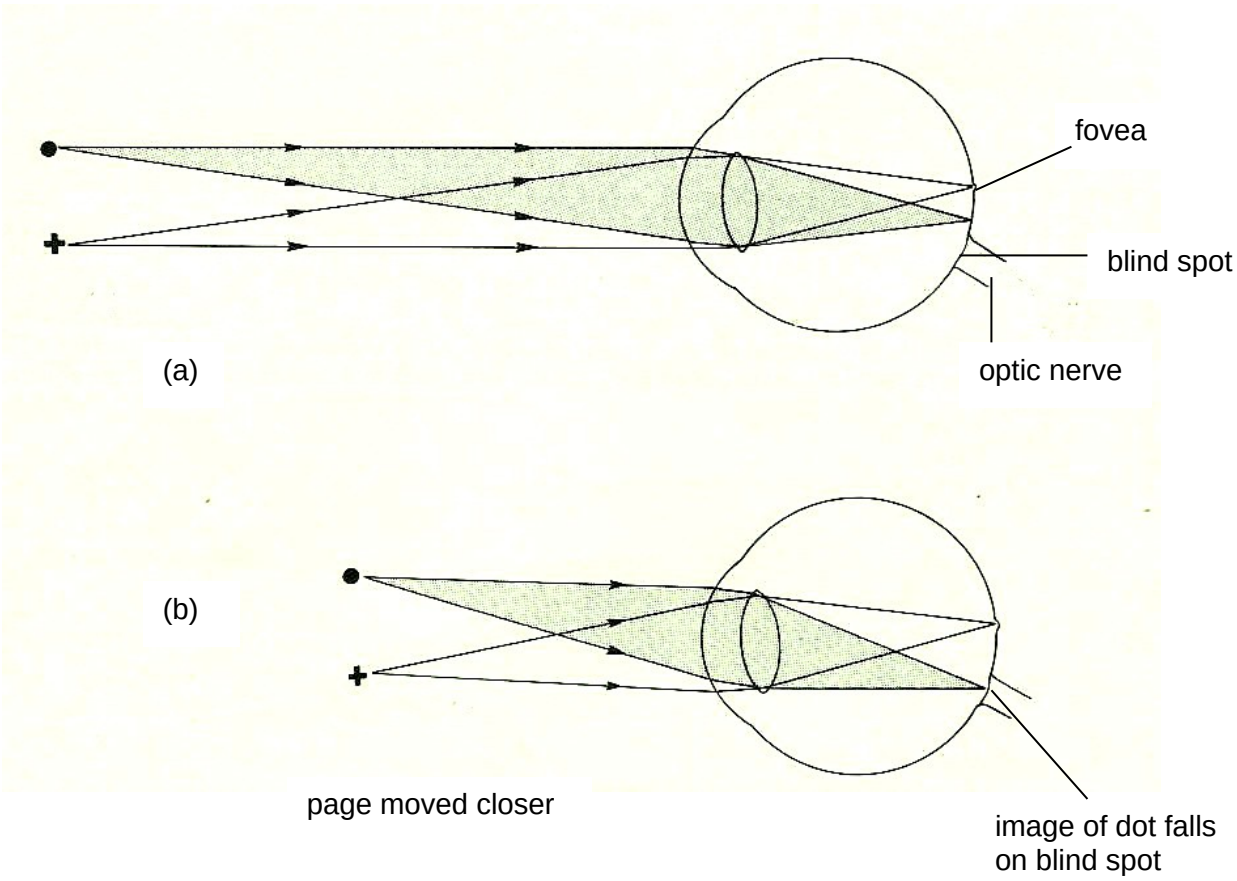


Fig. 1 Right eye in *horizontal* section